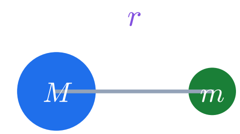
章末比例題解題圖：先把質量與幾何尺度畫成正確倍率

A｜Q1：質量一次、距離平方

$F \propto Mm/r^2$



$F'/F = 3/5^2 = 3/25$

B｜Q2：表面距離就是半徑

$g'/g_E = 4/(2^2) = 1$

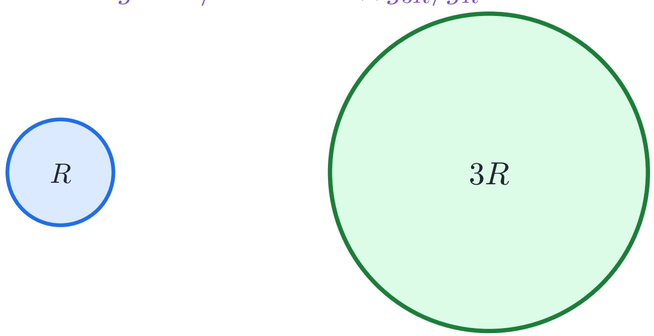
$g = GM/R^2$

C｜Q3：同密度球體的半徑倍率

同密度：$M \propto R^3$

$M_{3R}/M_R = 3^3 = 27$

$g \propto M/R^2 \propto R$，故 $g_{3R}/g_R = 3$



D｜Q9：M, R 同倍率時 GM/R 不變

$GM'/R' = G(2M)/(2R) = GM/R$

$v_{\text{orb}} = \sqrt{GM/R}$，$v_{\text{esc}} = \sqrt{2GM/R}$

兩天體的兩種速率相同，且 $v_{\text{esc}}/v_{\text{orb}} = \sqrt{2}$